Year 13 Maths - Hypothesis testing

Similar questions - Pearson's correlation coefficient

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Critical values of Pearson's product-moment correlation coefficient r (one-tailed test at the stated level):

n	5% (1-tail)	2.5%	1%	0.5%
9	0.5822	0.6664	0.7498	0.7977
10	0.5494	0.6319	0.7155	0.7646
12	0.4973	0.5760	0.6581	0.7079

A teacher records the hours x spent revising and the test mark y for 12 students and finds $r = 0.612$.

(a) Test, at the 5% significance level, whether there is positive correlation between x and y. **[4]**

(b) Explain why the same value of r would not be significant in a two-tailed test at the 1% level. **[2]**

(c) A student says: "This proves that revising more causes higher marks." Comment. **[2]**

2 A cafe owner records, on 9 days, the temperature T and the number S of hot drinks sold. The product-moment correlation coefficient is $r = -0.652$. Using the table above, test at the 5% significance level (two-tailed) whether there is any linear correlation between T and S, and state the conclusion in context. **[5]**

Answers

Question 1

(a) $H_0 : \rho = 0$, $H_1 : \rho > 0$. $n = 12$, 5% one-tailed critical value 0.4973. $0.612 > 0.4973$, so reject H_0: there is evidence of positive linear correlation between revision time and mark

(b) The two-tailed 1% critical value for $n = 12$ is 0.7079 (the 0.5% one-tail column). $0.612 < 0.7079$, so it is not significant

(c) Correlation does not imply causation: other factors (ability, prior attainment) may explain both

Question 2

$H_0 : \rho = 0$, $H_1 : \rho \neq 0$. For $n = 9$ a two-tailed 5% test uses the 2.5% column: 0.6664. $|-0.652| = 0.652 < 0.6664$, so do not reject H_0: there is insufficient evidence at the 5% level of linear correlation between temperature and hot drinks sold